# Option Pricing Using Binomial Model and Cox-Ross-Rubinstein (CRR) Model

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# Outline


1. Option pricing using delta hedging
2. Option pricing using underlying and risk free asset
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# What is an equity option ?

**Definition (Equity option)**

An equity option is a contract which conveys to its holder the right, but not the obligation, to buy or sell shares of the underlying security at a specified price on or before a Specified date. This right is granted by the seller of the option.

- Buy = Call option. Sell = Put option
- On/before: American. Only on expiration: European
- Specified price = Strike or exercise price
- Specified date = Maturity or expiration date
- Buyer = holder = long position
- Seller = writer = short position

**Example (European-style put option)**

An European-style XYZ May 60 put entitles the buyer to sell 100 shares of XYZ Corp. common stock at $60 per share **at the expiration date** in May.

# One-Period Binomial Model

**Why hedging ?**

- Buying a stock is a risky investment.
- Buying a call option on that stock is also risky.
- Combining the stock with the option can produce an investment that is risk free.

## Delta hedging

The number of units of the stock required per option is known as the $\Delta$ (Delta) of the stock and taking these positions to create a risk-free investment is known as *Delta Hedging*.

### Definition (Arbitrage)

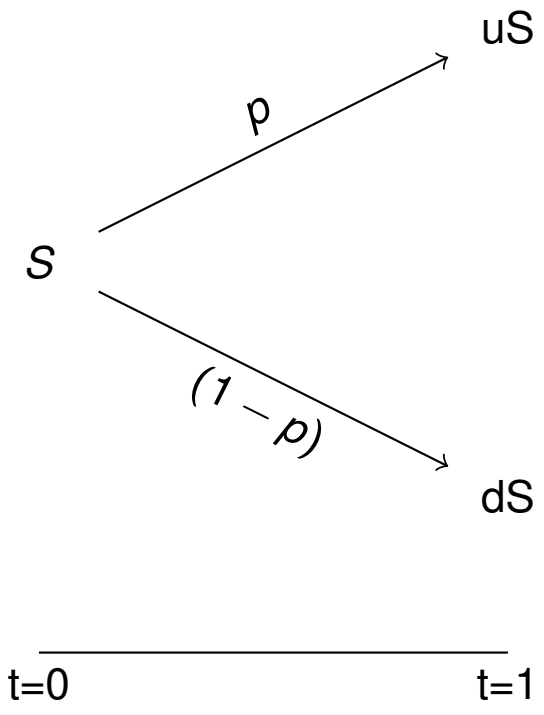
The simultaneous buying and selling of a security at two different prices in two different markets, resulting in profits without risk. Perfectly efficient markets present no arbitrage opportunities. Perfectly efficient markets seldom exist, but, arbitrage opportunities are often precluded because of transactions costs.

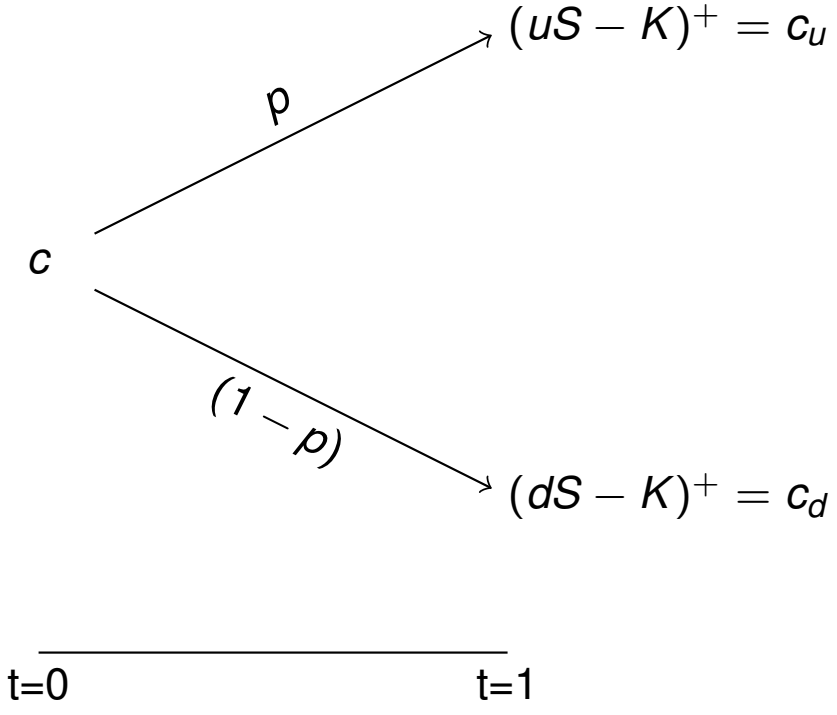
### Example

As a hypothetical example, if Infosys stock trades at Rs 790 on the National Stock Exchange (NSE) and at Rs 800 on the Bombay Stock Exchange (BSE), an investor could guarantee a profit by purchasing the stock on the NSE and simultaneously selling the same amount of stock on the BSE.

# Payoffs from equity and call option

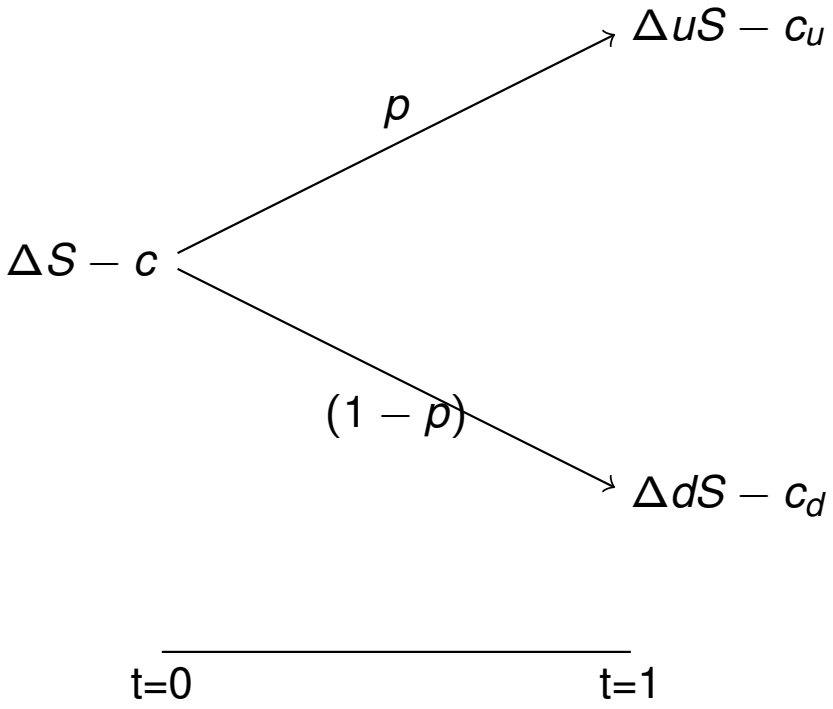
- Consider an equity with price $S$ at time 0.
- Let the price will be $uS$ under up movement and $dS$ under down movement.
- Suppose there is an European call option with strike price $K$ with expiry at time 1.
- Note that call option payoff will be $(uS - K)^+$ in up state and $(dS - K)^+$ in down state at time 1.



# Portfolio of writing one call and buying $\Delta$ share of equity

Suppose that we buy $\Delta$ units of the stock and sell one call at time $t = 0$. The portfolio value will be $\Delta uS - (uS - K)^+$ if equity price moves up and $\Delta dS - (dS - K)^+$ if equity price moves down at time $t = 1$.



- To make this portfolio risk free, it is necessary to choose $\Delta uS - c_u = \Delta dS - c_d$, i.e. $\Delta = \frac{c_u - c_d}{S(u-d)}$.
- Further, if $r$ is the interest rate for one period, we should have

$$\Delta S - c = \frac{\Delta dS - c_d}{1+r} = \frac{\Delta uS - c_u}{1+r},$$

which implies after substitution of value of $\Delta$

$$\begin{aligned} c &= \frac{1}{1+r}\left[\frac{1+r-d}{u-d}c_u + \frac{u-1-r}{u-d}c_d\right] \\ &= \frac{1}{1+r}(\tilde{p}c_u + (1-\tilde{p})c_d), \end{aligned} \tag{1}$$

where $\tilde{p} = \frac{(1+r-d)}{(u-d)}$.

- If $c$ is not so, there will be an arbitrage opportunity.
- Interestingly in option price formula the probability term $p$ doesn't appear.

# An example

- Suppose the current price of the underlying stock is 100 and at the end of three months it has either risen to 175 or has fallen to 75.
- Consider an at-the-money call option on this stock with a strike price of 100 with expiry after 3 months.
- Suppose for 3-months risk free rate of interest is 25% (or 100% annual).

**Find $u$, $d$, $r$ and $\tilde{p}$ ?**

$u = 1.75, d = .75, r = .25, \tilde{p} = 1/2.$

**What are the values of $c_u$ and $c_d$ ?**

$c_u = (uS - K)^+ = \max(uS - K, 0) = \max(75, 0) = 75.$
$c_d = (dS - K)^+ = \max(dS - K, 0) = \max(-25, 0) = 0.$

**What is $\Delta$ ?**

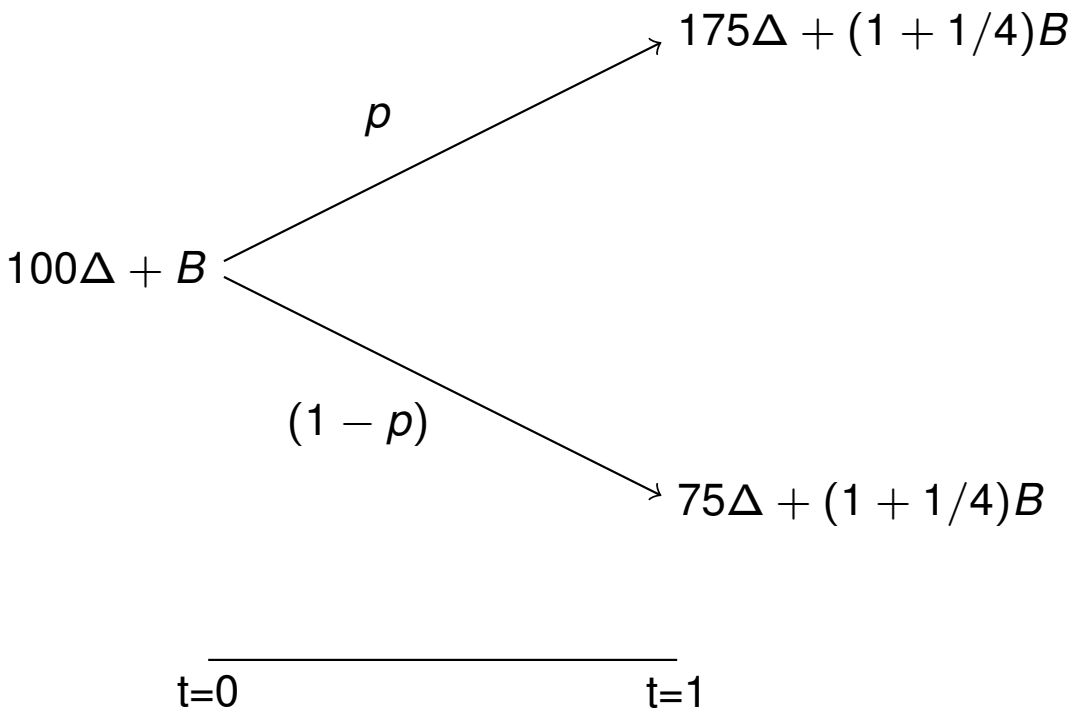
$\Delta = \frac{c_u - c_d}{S(u-d)} = 3/4.$

**What is option price $c$ ?**

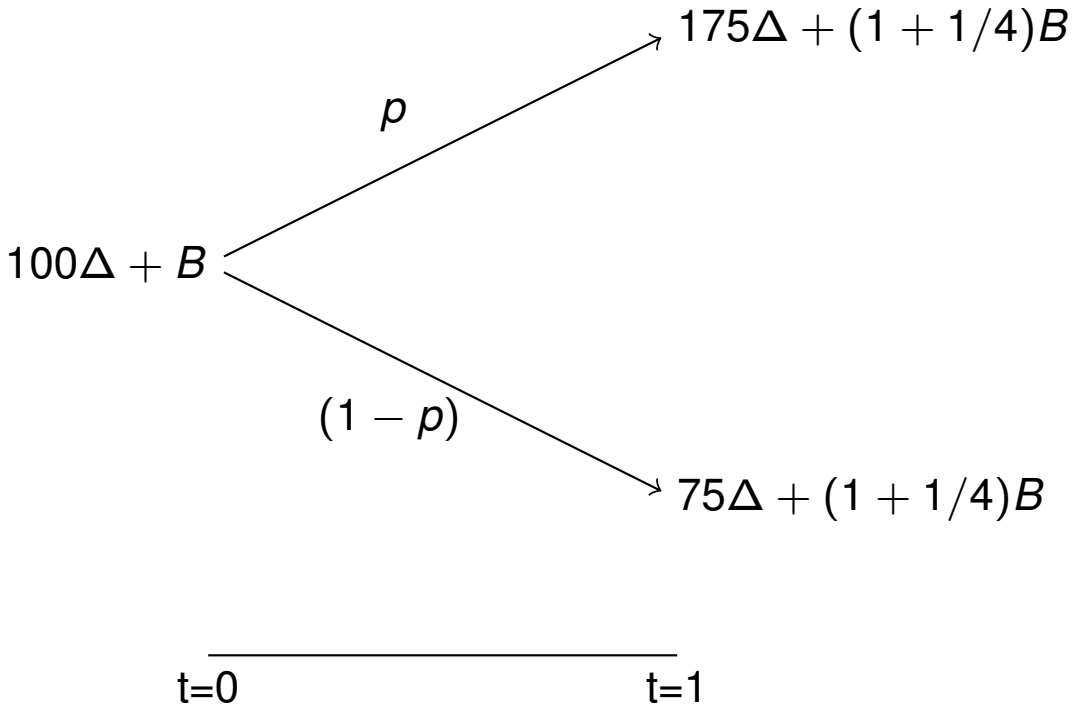
$c = \frac{1}{1+r}(\tilde{p}c_u + (1 - \tilde{p})c_d) = 30.$

# Example

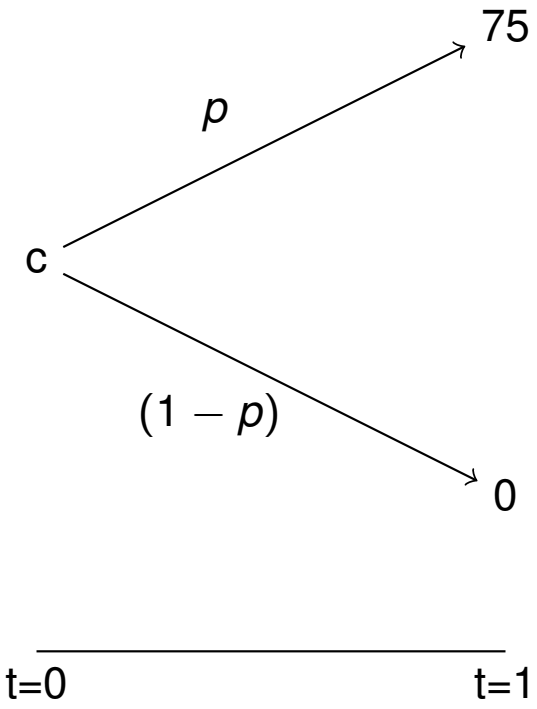
- Consider again the same example, where the price of the underlying stock is 100 initially and at the end of 3 months it has either risen to 175 or fallen to 75.
- Consider an European call option of strike price 100.
- the payoff of the call option in up state is 75 and payoff in the down state is 0 and the option will not be exercised.
- Consider buying $\Delta$ unit of the stock and investing $B$ units of fund at risk free interest rate of 25% for 3-months (or 100% annualized).



# Replicating portfolio




Cash-flows from portfolio




Cash-flows from call option

# Solving for $\Delta$ and $B$

- To replicate the cash-flows from option, we should have $175\Delta + (1 + 1/4)B = 75$ and $75\Delta + (1 + 1/4)B = 0$, which implies $\Delta = 3/4$ and $B = -45$.
- Thus option can be replicated by borrowing 45 and buying 3/4 units of the stock.
- The value of this portfolio at time 0 is $100 \times 3/4 - 45 = 30$.
- Due to no-arbitrage principle the portfolio that replicates the call option will have same price as the call itself that is $c = \Delta S + B$.
- Hence the price of the call is $c = 30$ which is same as the previous case.

# If $c \neq \Delta S + B$, we have arbitrage

Suppose $c = 25$, then we consider the following portfolio at $t = 0$. This means $\Delta S + B > c$.

- Short sell $3/4$ shares and generate 75.
- Buy one call option with 25.
- Invest the remaining 50 in bank. This is a zero investment portfolio.

If there is an up movement, at maturity, we will have the following:

- Exercise the call option and receive one share by paying Rs 100. Return 3/4 shares which were short sell. Sell the remaining 1/4 shares with market price Rs $175/4 = 43.75$.
- The investment in the bank will be $50(1 + .25) = 62.5$.
- Total payoff at maturity = $62.5 + 43.75 - 100 = 6.25$.

# If $c \neq \Delta S + B$, we have arbitrage contd...

If there is a down movement, at maturity, we will have

- The call option is worthless.
- Buy the 3/4 shares from the market with amount $75 \times 3/4 = 56.25$ and return to the person from whom the equity was borrowed for short sell.
- The investment in the bank will be $50(1 + .25) = 62.5$.
- Total payoff at maturity = $62.5 - 56.25 = 6.25$.

# If $c > \Delta S + B$, we have arbitrage

Suppose $c = 40$, then we consider the following portfolio at $t = 0$. This means $c > \Delta S + B$.

- Sell the call option and receive 40.
- Buy 3/4 shares by paying Rs 75.
- Borrow the remaining 35 from the bank with interest rate 0.25 for the period. This is a zero investment portfolio.

If there is an up movement, at maturity, we will have the following:

- The call option will be exercised and we give the 3/4 share we bought at strike price 100. Further, we buy the remaining 1/4 shares from the market with price 175/4 = 43.75. In this process, we recive the strike price of the option, which is 100.
- The loan from the bank will be $35(1 + .25) = 43.75$.
- Total payoff at maturity = $100 - 43.75 - 43.75 = 12.5$, which is an arbitrage opportunity.
- Similarly for the down movement.

- From earlier discussion, we have

$$c = \frac{1}{1+r}(\tilde{p}c_u + (1-\tilde{p})c_d),$$

where $\tilde{p} = \frac{(1+r-d)}{(u-d)}$.

- Here $\tilde{p}$ can be interpreted as probability if $d < 1 + r < u$.
- Here the value of the call option is the present value of the weighted average of the call at maturity.
- The value of the call is present value of $\tilde{p}c_u + (1-\tilde{p})c_d$ and is valued as if this payoff was risk-free.

### Risk-neutral probability

The risk-neutral probability is that probability under which expected returns on all the model's assets are the same.

# Dividend yield and continuous compounding

### Dividend yield

Suppose the equity pays a dividend with dividend yield $q$ for the period. Then we calculate the risk-neutral probabilities as follows:

$$\tilde{p}uS + (1 - \tilde{p})dS = (1 + r - q)S,$$

which implies

$$\tilde{p} = \frac{1 + (r - q) - d}{u - d}.$$

### Continuous compounding

For continuous compounding, the risk neutral probability become for no-dividend

$$\tilde{p} = \frac{(e^r - d)}{(u - d)}.$$

### Continuous compounding and dividend

If there is a dividend, for continuous compounding, the risk neutral probability is

$$\tilde{p} = \frac{(e^{r-q} - d)}{(u - d)}.$$

In a one period Binomial model, we always have $d < 1 + r < u$, otherwise there will be an arbitrage opportunity.

**Proof.**

- Suppose $d \geq 1 + r$, this mean in up and down moment both equity return is higher than the risk free rate.
- Thus, we buy the equity after borrowing $S_0$ from bank.
- After one period total payoff will be more than or equal to $dS_0 - (1 + r)S_0$, which is a non-negative quantity.
- Hence $d \geq 1 + r$ leads to an arbitrage opportunity, which is not possible in efficient markets.
- Similarly, by doing short selling, we prove that $1 + r \geq u$ is not possible.

**Remark**

*The no-arbitrage condition for continuous compounding for a dividend paying stock will be*

$$d < e^{r-q} < u.$$

**Example**

The current price of a certain non-dividend-paying stock is \$100 per share. Suppose the price of this stock is modeled at the end of a quarter year using a one-period binomial tree under the assumption that the stock price can either increase by 4%, or decrease by 2%.The continuously compounded risk-free interest rate is 3%. What is the price of a three-month, at-the-money European call option on the above stock ?

**Solution:** We have the following

- $r = 0.03/4 = 0.0075$.
- $u = 1.04$ and $d = 0.98$.
- $S = 100$ and $K = 100$.
- $c_u = 4$ and $c_d = 0$.
- The risk neutral probability is $\tilde{p} = \frac{e^{.0075} - 0.98}{1.04 - 0.98} = 0.4588$.
- Thus option price $c = e^{-.0075}(0.4588 \times 4) = 1.82149$.

### Equivalent Martingale Measure (EMM)

A common method for pricing an asset is to use a risk-neutral or an equivalent martingale measure (EMM). The EMM is convenient because all asset prices are simply an expectation of the payoff.

### Definition (EMM)

The probability measure $\mathbb{Q}$ is an EMM of $\mathbb{P}$ if $S_t$ is a $\mathbb{Q}$-martingale, that is

$$\mathbb{E}^Q S_T = S_0$$

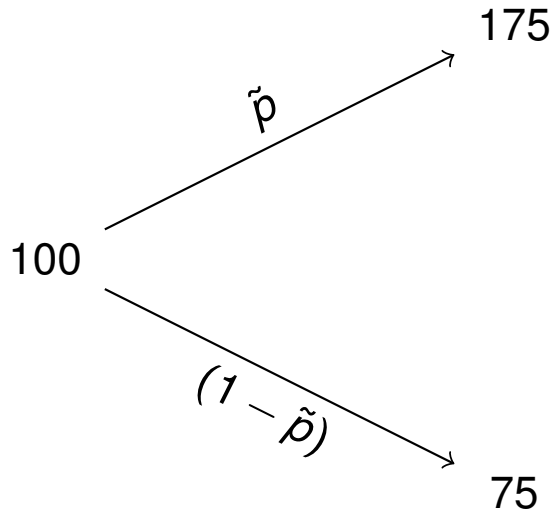
and $\mathbb{Q}$ is equivalent to $\mathbb{P}$,

$$\mathbb{P}(\omega) > 0 \iff \mathbb{Q}(\omega) > 0$$

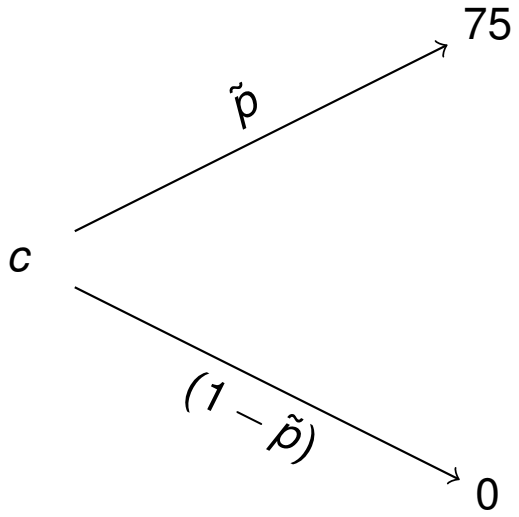
for all elementary events $\omega \in \Omega$.

Under risk neutral probabilities the discounted value of expected cash flows is equal to the actual value of the underlying asset.

**Example (Revisiting the example)**




Stock cash-flows



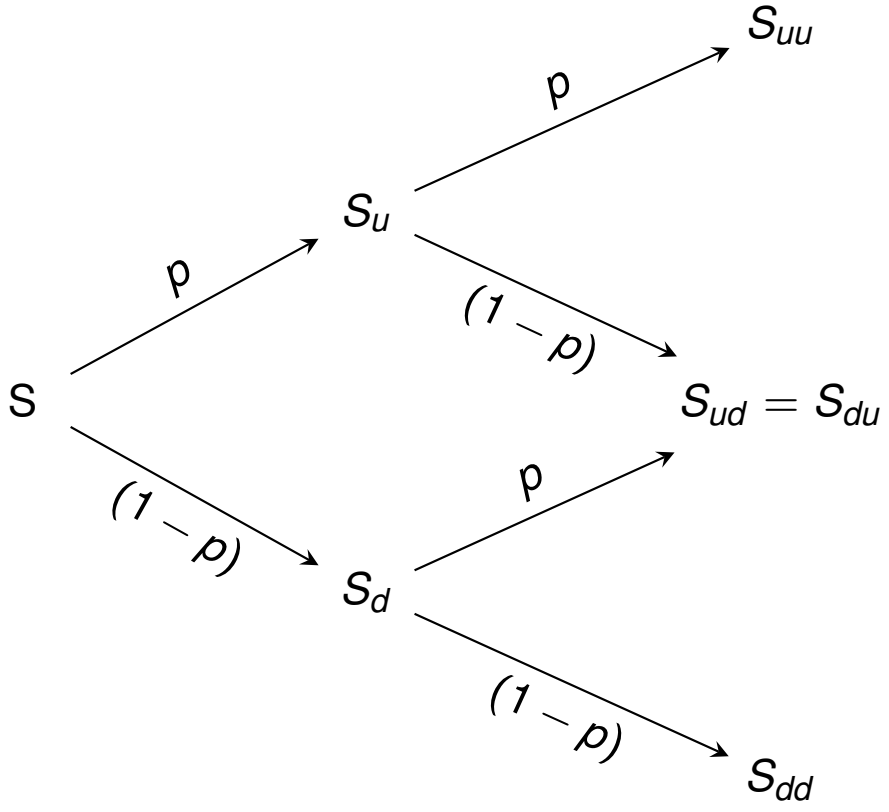

Option cash-flows

- Under risk neutral probabilities the expected value of the stock at the end of period is $175\tilde{p} + 75(1 - \tilde{p})$.
- So the expected present value is $\frac{175\tilde{p}+75(1-\tilde{p})}{1+1/4}$.
- This is equal to the actual value of the stock that is 100, which implies $\tilde{p} = 1/2$.
- Thus option value is $\frac{75*1/2+0*1/2}{1+1/4} = 30$.

# Two-Period Binomial Model

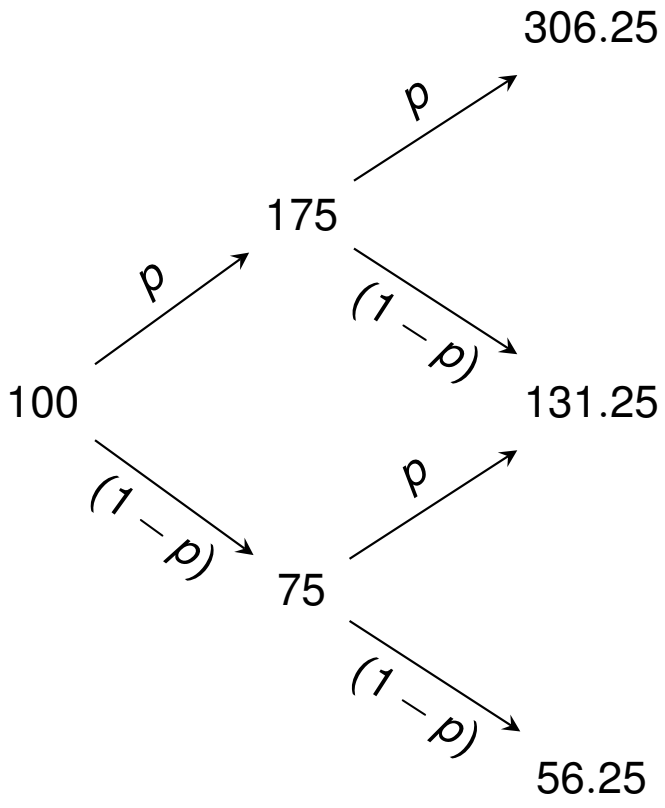
# Two-period Binomial model

- The one-period model is extended by a two-period model.
- The value of the stock after two periods when both movements are up will be $u^2 S = S_{uu}$ and it will be $d^2 S = S_{dd}$ if both movements are down.
- If the stock first went up and then went down or it first went down and then went up, it will be $udS = S_{ud} = S_{du}$.

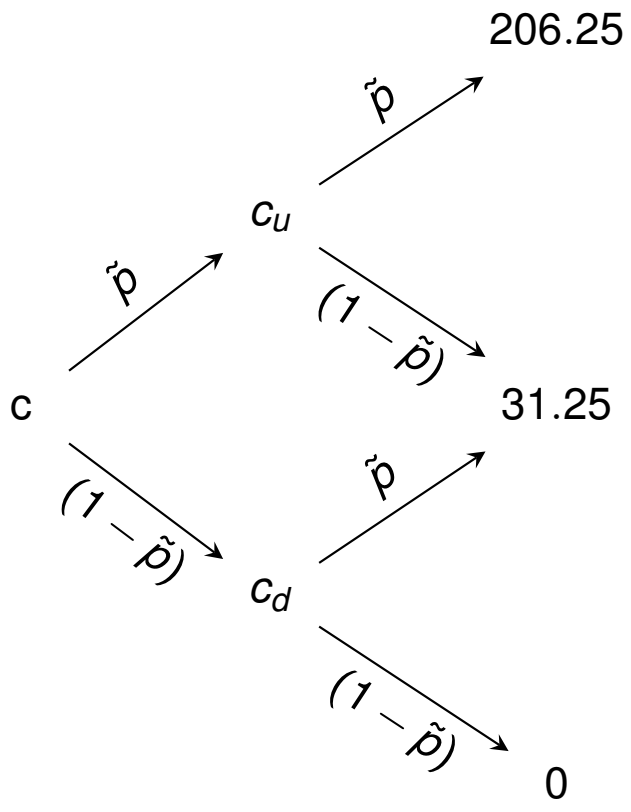


# Example

- Consider the earlier discussed example.
- We have $u = 1.75$, $d = 0.75$, $S = 100$, $K = 100$, $r = 100\%$ or $r = 1/4$ for 3-months.

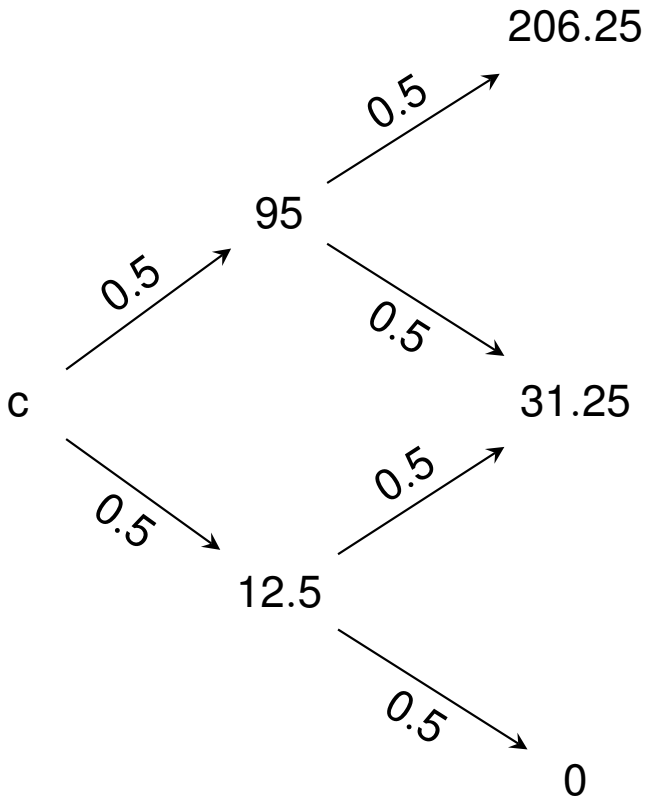


# Finding value of call at intermediate steps



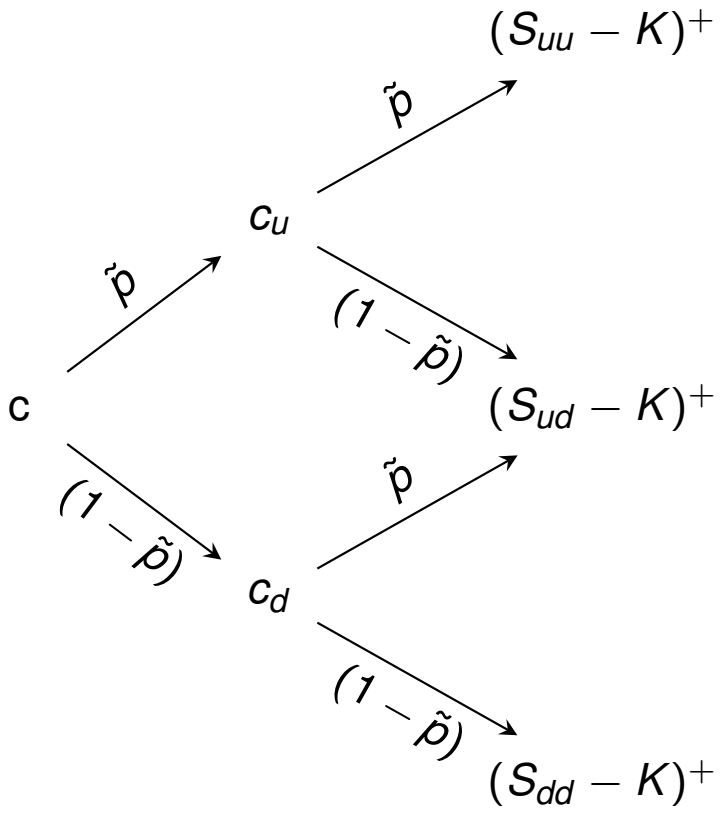

- The value of the call at the intermediate nodes can be found by using delta hedging method or risk free rate and underlying asset method or by using risk neutral valuation method.
- The risk neutral method is simple because the node probabilities are same for different periods.
- Hence $c_u = \frac{0.5\times 206.25+0.5\times 31.25}{1+1/4} = 95$
- $c_d = \frac{0.5\times 31.25+0.5\times 0}{1+1/4} = 12.5$

# Finding value of call at time 0




- Again using the similar steps as earlier, we can find the no-arbitrage price of call option.
- $c = \frac{0.5 \times 95 + 0.5 \times 12.5}{1+1/4} = 43.$
- Alternatively, $c = \frac{0.5 \times 0.5 \times 206.25 + 2 \times 0.5 \times 0.5 \times 31.25 + 0.5 \times 0.5 \times 0}{(1+1/4)^2} = 43.$

# Converting into formula




$$c = \frac{\tilde{p}^2(S_{uu} - K)^+ + 2 \times \tilde{p}(1 - \tilde{p})(S_{ud} - K)^+ + (1 - \tilde{p})^2(S_{dd} - K)^+}{(1 + r)^2}$$

# n-Period Binomial Model

### Revisiting Two-period Binomial model

We know that

$$
\begin{aligned}
c &= \frac{\tilde{p}^2(S_{uu}-K)^+ + 2\times\tilde{p}(1-\tilde{p})(S_{ud}-K)^+ + (1-\tilde{p})^2(S_{dd}-K)^+}{(1+r)^2} \\
&= \frac{\binom{2}{2}\tilde{p}^2(1-\tilde{p})^0(S_{uu}-K)^+ + \binom{2}{1}\times\tilde{p}(1-\tilde{p})(S_{ud}-K)^+ + \binom{2}{0}\tilde{p}^0(1-\tilde{p})^2(S_{dd}-K)^+}{(1+r)^2} \\
&= \frac{1}{(1+r)^2}\sum_{i=0}^{2}\binom{2}{i}\tilde{p}^i(1-\tilde{p})^{2-i}\left(u^i d^{(2-i)}S-K\right)^+
\end{aligned}
$$

### n-period call option price

Thus the two-period option price formula can be easily generalized to n-period formula

$$
c = \frac{1}{(1+r)^n}\sum_{i=0}^{n}\binom{n}{i}\tilde{p}^i(1-\tilde{p})^{n-i}\left(u^i d^{(n-i)}S-K\right)^+,
$$

where $\tilde{p} = \frac{1+r-d}{u-d}$.

# Cox-Ross-Rubinstein (CRR) model

### Model setup

- CRR model is based on the assumption of a binomial model.
- It can be interpreted as a numerical method to solve the Black-Scholes equation.
- we discretize time and consider points in time such that $t_0 = 0, t_1 = \Delta, t_2 = 2\Delta, ..., t_n = n\Delta = T$ with $\Delta = \frac{T}{n}$.
- For large $n$, the option price from CRR model and Black-Scholes model will be approximately same.
- By choosing

$$u = e^{\sigma\sqrt{\Delta}}, \quad d = \frac{1}{u}, \quad \tilde{p} = \frac{(e^{r\Delta} - d)}{(u - d)} \approx \frac{1}{2} + \frac{1}{2}(r - \frac{1}{2}\sigma^2)\frac{\sqrt{\Delta}}{\sigma},$$

in Binomial model, we get the CRR model, where $\sigma$ is volatility term in the model.

# A working example

**Table:** Call option features

| | | |
|---|---|---|
| Current stock price | $S$ | 230.00 |
| Exercise price | $K$ | 210.00 |
| Time to maturity | $T$ | 0.50 |
| Volatility | $\sigma$ | 0.25 |
| Risk free rate | $r$ | 0.04545 |
| Cost of carry | $b$ | 0.04545 |
| Time steps | $n$ | 5 |
| Option type | | European call |

- We have $\Delta = T/n = 0.1$, $u = e^{\sigma\sqrt{\Delta}} = e^{0.25\sqrt{0.1}} = 1.0823$, and $d = 1/u = 0.9234$.
- Given the current stock price $S = 230$ the stock can either increase to $S_u = uS = 248.92$ or decrease to $S_d = dS = S/u = 212.52$.
- After the second time step, we have $S_{uu} = uS_u = 269.40$ , $S_{ud} = S_{du} = 230$, and $S_{dd} = 196.36$ and so on. At maturity, after 5 time steps, the stock price can take the following six values $S_{uuuuu} = u^5 S = 341.51$, $S_{uuuud} = u^4 dS = u^3 S = 291.56$, $\ldots$, $S_{ddddd} = S/u^5 = 154.90$.

# Equity price evolution under CRR model

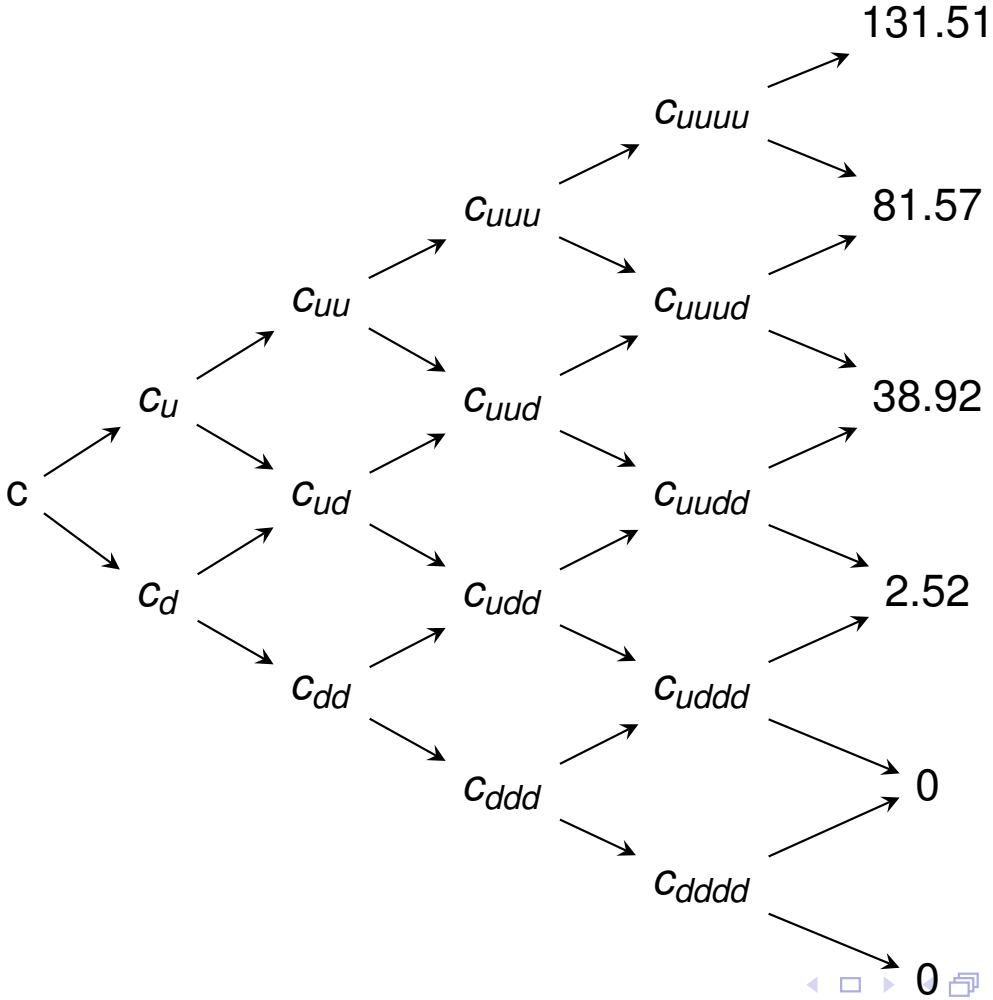
| | | | | | |
|---|---|---|---|---|---|
| | | | | | 341.51 |
| | | | | 315.55 | |
| | | | 291.56 | | 291.57 |
| | | 269.40 | | 269.4 | |
| | 248.92 | | 248.92 | | 248.92 |
| 230.00 | | 230.00 | | 230.00 | |
| | 212.52 | | 212.52 | | 212.52 |
| | | 196.36 | | 196.36 | |
| | | | 181.44 | | 181.44 |
| | | | | 167.65 | |
| | | | | | 154.90 |

# Option value at different nodes

We have $\tilde{p} = 0.50898$ and $r = 0.04545$, hence

$$c_{uuuu} = \frac{0.50898 \times 131.51 + (1 - 0.50898) \times 81.57}{e^{0.04545 \times 0.1}}$$

$$\approx \frac{0.50898 \times 131.51 + (1 - 0.50898) \times 81.57}{1 + 0.04545 \times 0.1} = 106.50$$



# Dividend paying stock

- If stock dividend yield is $q$, then

$$u = e^{\sigma\sqrt{\Delta}}, \quad d = \frac{1}{u}, \tilde{p} = \frac{(e^{(r-q)\Delta} - d)}{(u-d)} \approx \frac{1}{2} + \frac{1}{2}((r-q) - \frac{1}{2}\sigma^2)\frac{\sqrt{\Delta}}{\sigma}$$
$$= \frac{1}{2} + \frac{1}{2}(b - \frac{1}{2}\sigma^2)\frac{\sqrt{\Delta}}{\sigma},$$

  where $b$ is called cost of carry for equity.
- Note that, we will still discount by $e^{r\Delta} \approx (1 + r\Delta)$ for small $\Delta$.

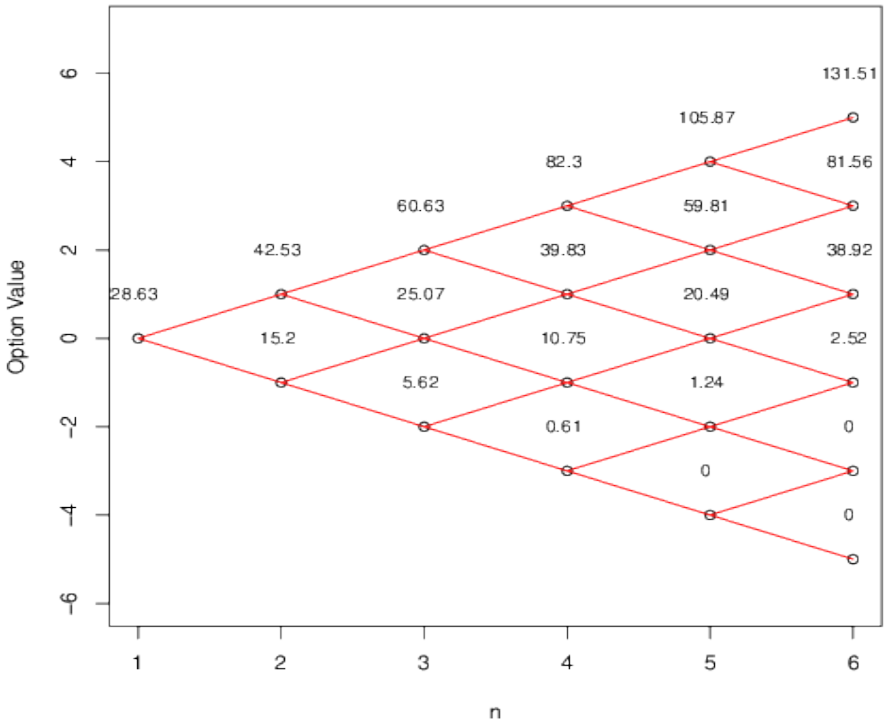
**Table:** Call option features

| | | |
|---|---|---|
| Current stock price | $S$ | 230.00 |
| Exercise price | $K$ | 210.00 |
| Time to maturity | $T$ | 0.50 |
| Volatility | $\sigma$ | 0.25 |
| Risk free rate | $r$ | 0.04545 |
| Cost of carry | $b$ | 0.02545 |
| Time steps | $n$ | 5 |
| Option type | | European call |

# Option value at different nodes

We have $\tilde{p} = 0.49633$ and $r = 0.04545$, hence

$$c_{uuuu} = \frac{0.49633 \times 131.51 + (1 - 0.49633) \times 81.57}{1 + 0.04545 \times 0.1} = 105.87.$$




```
library(fOptions)
CRRTree = BinomialTreeOption(TypeFlag = "ce", S = 230, X = 210,Time = .5, r =
0.04545, b = 0.02545, sigma = 0.25, n = 5);
BinomialTreePlot(CRRTree, dy = 1, cex = 0.8, ylim = c(-6, 7),xlab = "n", ylab = "Option
Value")
```

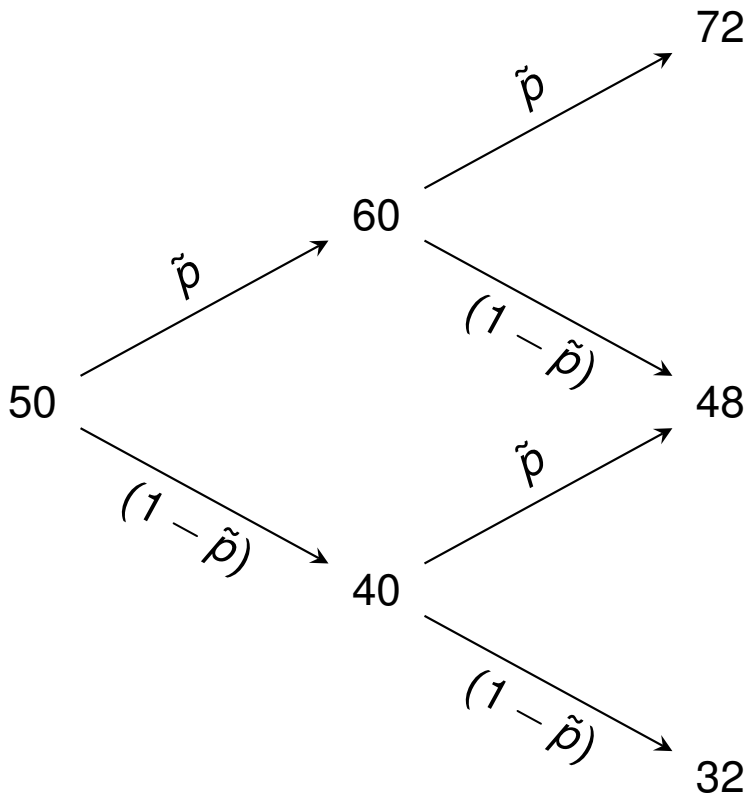
# Infosys Option Price

```
library(quantmod)
library(fOptions)
INFY = getSymbols('INFY.NS',src='yahoo', from = "2017-01-01")
INFY = INFY.NS[,6]
INFY.rets = diff(log(INFY))[-1]
INFY.sd = sd(INFY.rets)*sqrt(252)
S = as.numeric(INFY.NS[nrow(INFY.NS),4])
X= 1160; T = 28/365; r = .0631; b= r; sigma = INFY.sd; n=100
CRRBinomialTreeOption(TypeFlag = "ce", S = S, X = X,Time = T, r = r, b = b, sigma =
sigma, n = n)
```

# European put option under Binomial model

**Example**

Suppose we have a 2 year European put with $K = Rs\ 52$ and $S = Rs\ 50$. Suppose we consider 2 time steps each of duration 1 year, in which the stock goes up or down by 20%. Suppose risk free rate $r = 0.05$. We have

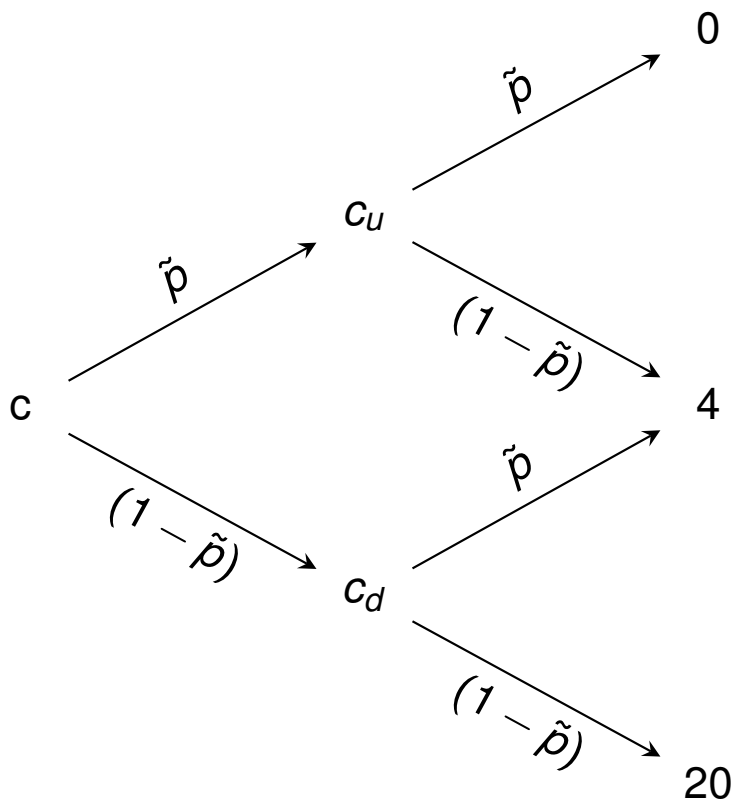


# Example of European put option contd...

The risk neutral probability

$$\tilde{p} = \frac{1 + r - d}{u - d} = \frac{1 + .05 - 0.8}{1.2 - 0.8} = 0.625.$$

The put option payoff are as follows under the Binomial model




We get $c_u = 1.429$, $c_d = 9.524$ and **put option price** $c = 4.252$.

# American Options

# American option

- An **American style option** allows the buyer to exercise the option at any time up to the expiration time (maturity).
- However an **European style option** can only be exercised at maturity.
- American options could be call option or put options.
- Although it might seem that, because of its additional flexibility, the **American style call option** would be worth more, however the two style **call options** without any dividend on underlying asset have identical worth.
- The American and European style put options have different worth.

# American call worth is same as European call

**Proposition**

*For a non-dividend paying stock, one should not exercise an American call option before its expiration time $T$.*

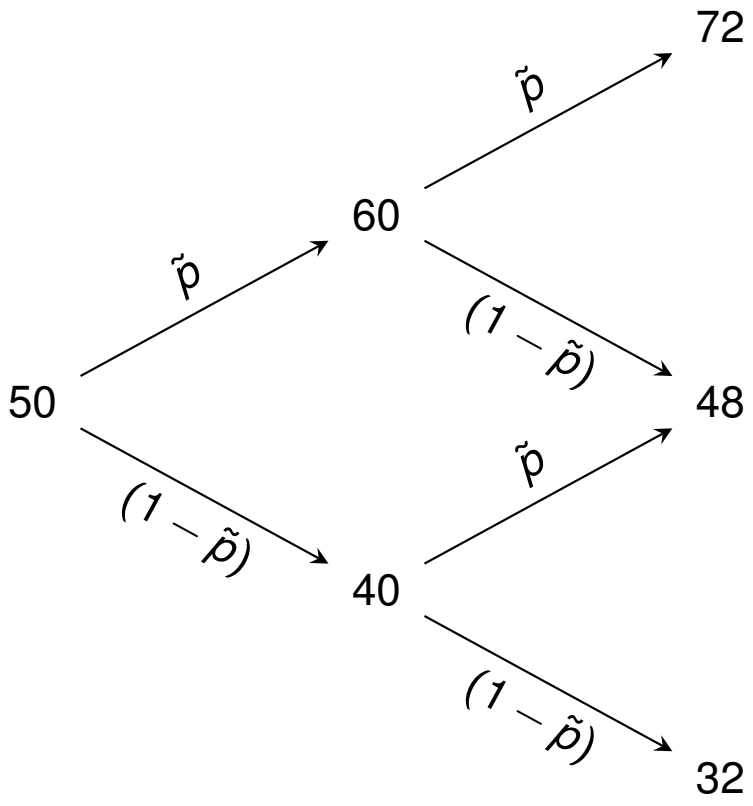
**Proof.**

- Suppose the current price of the stock is $S$ and we have the American call option.
- The time to expire of the option is $t$ from now.
- If we exercise the option at this moment, we will realize the amount $S - K$.
- However consider the following strategy instead of exercising the option.
  - (i) Keep the call option.
  - (ii) Short sell the stock and receive amount $S$ and put it in the bank which will give $Se^{rt}$ at maturity.
  - (iii) At maturity, we will have
  $$\begin{cases} Se^{rt} - K & \text{if } S_T \geq K \\ Se^{rt} - S_T & \text{if } S_T \leq K, \end{cases}$$
  which is larger than $(S - K)e^{rt}$ in both the cases and hence their is no benefit in exercising the American call option before maturity.
- Thus American call and European call options will have the same price for a non-dividend paying stock.

$\square$

**Example**

Suppose we have a 2 year American put option with $K = Rs\ 52$ and $S = Rs\ 50$. Suppose we consider 2 time steps each of duration 1 year, in which the stock goes up or down by 20%. Suppose risk free rate $r = 0.05$. We have

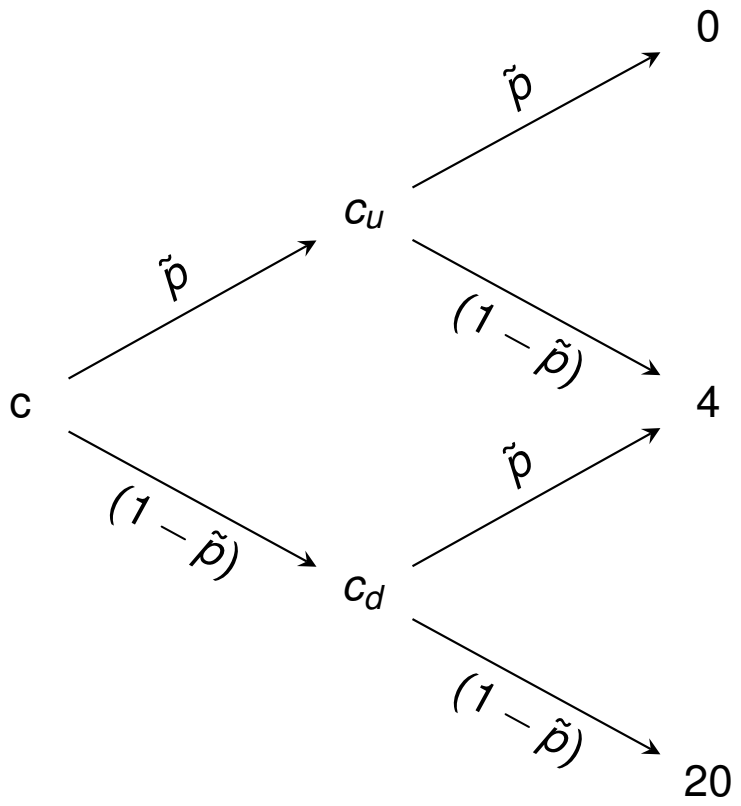


# Pricing American put option contd...

The risk neutral probability

$$\tilde{p} = \frac{1 + r - d}{u - d} = \frac{1 + .05 - 0.8}{1.2 - 0.8} = 0.625.$$

The put option payoff are as follows under the Binomial model



# Pricing American put contd...

- We have

$$\tilde{c}_u = \max\{K - uS, c_u\} = \max\left\{-8, \frac{0.625 \times 0 + 0.375 \times 4}{1.05}\right\} = \max\{-8, 1.429\} = 1.429.$$

- Further,

$$\tilde{c}_d = \max\{K - dS, c_d\} = \max\left\{12, \frac{0.625 \times 4 + 0.375 \times 20}{1.05}\right\} = \max\{12, 9.524\} = 12.$$

- Thus American put option price will be

$$\text{put price} = \max\left\{2, \frac{0.625 \times 1.429 + 0.375 \times 12}{1.05}\right\} = \max\{2, 5.136\} = 5.136.$$

- Note that American style put option price is higher than the European style put option price.

# Greeks under CRR Model

# Influential Greeks

## Greeks

The option price sensitivities to different input parameters are called "Greeks". Greeks are mostly used for hedging purposes. The most important Greek is the Delta. Hedging an option with a Delta is called delta hedging.

1. The **Delta** of a derivative security denoted by $\Delta$ is defined as the rate of change of its price with respect to the price of the underlying asset. If the option price is $C$, then $\Delta = \frac{\partial C}{\partial S}$.

2. The second derivative of the call option with respect to the price of the stock is called the **Gamma** of the call option and is given by $\Gamma = \frac{\partial^2 C}{\partial S^2}$.

3. The rate of change of the price of the call option with respect to time to maturity is called **Theta** and denoted by $\Theta = \frac{\partial C}{\partial T}$.

# Estimating Greeks under CRR model

The prices look like the following under two period CRR model

$$
\begin{tikzcd}
 & & S_{uu} \\
 & S_u \arrow[ur] \arrow[dr] & \\
S \arrow[ur] \arrow[dr] & & S_{ud} = S_{du} = S \\
 & S_d \arrow[ur] \arrow[dr] & \\
 & & S_{dd}
\end{tikzcd}
$$

The option price at different nodes look like the following

$$
\begin{tikzcd}
 & & c_{uu} \\
 & c_u \arrow[ur] \arrow[dr] & \\
c \arrow[ur] \arrow[dr] & & c_{ud} \\
 & c_d \arrow[ur] \arrow[dr] & \\
 & & c_{dd}
\end{tikzcd}
$$

1. The $\Delta$ is given by
$$\Delta = \frac{c_u - c_d}{S_u - S_d} = \frac{c_u - c_d}{S(u-d)}.$$
2. We calculate $\Gamma$ as follows:
$$\Gamma_2 = \frac{c_{uu} - c_{ud}}{S_{uu} - S}, \qquad \Gamma_1 = \frac{c_{ud} - c_{dd}}{S - S_{dd}},$$
and the change in the underlying is $h = \frac{1}{2}(S_{uu} + S) - \frac{1}{2}(S_{dd} + S) = \frac{1}{2}(S_{uu} - S_{dd})$. The *Gamma* is
$$\Gamma = \frac{\Gamma_2 - \Gamma_1}{h}.$$
3. The $\Theta$ is given by
$$\Theta = \frac{c_{ud} - c}{2\Delta t}.$$

# Find the Greeks $\Delta$, $\Gamma$ and $\Theta$

**Table:** Call option features

| | | |
|---|---|---|
| Current stock price | $S$ | 230.00 |
| Exercise price | $K$ | 210.00 |
| Time to maturity | $T$ | 0.20 |
| Volatility | $\sigma$ | 0.25 |
| Risk free rate | $r$ | 0.05 |
| Cost of carry | $b$ | 0.05 |
| Time steps | $n$ | 2 |
| Option type | | European call |

- We have $\Delta t = T/n = 0.1$, $u = e^{\sigma\sqrt{\Delta}} = e^{0.25\sqrt{0.1}} = 1.0823$, and $d = 1/u = 0.9234$.
- Given the current stock price $S = 230$ the stock can either increase to $S_u = uS = 248.92$ or decrease to $S_d = dS = S/u = 212.52$.
- After the second time step, we have $S_{uu} = uS_u = 269.40$ , $S_{ud} = S_{du} = 230$, and $S_{dd} = 196.36$.
- Further $\tilde{p} = 0.50898$.

# The Tree under CRR model

The prices look like the following under two period CRR model

$$
\begin{tikzcd}
 & & 268.40 \\
 & 248.92 \arrow[ur] \arrow[dr] & \\
230 \arrow[ur] \arrow[dr] & & 230 \\
 & 212.52 \arrow[ur] \arrow[dr] & \\
 & & 196.36
\end{tikzcd}
$$

The option price at different nodes look like the following

$$
\begin{tikzcd}
 & & 58.40 \\
 & c_u \arrow[ur] \arrow[dr] & \\
c \arrow[ur] \arrow[dr] & & 20 \\
 & c_d \arrow[ur] \arrow[dr] & \\
 & & 0
\end{tikzcd}
$$

We have

$$c_u = \frac{58.40 \times 0.50898 + 20 \times (1 - 0.50898)}{1 + .05 \times 0.1} = 39.35.$$

Similarly

$$c_d = \frac{20 \times 0.50898 + 0 \times (1 - 0.50898)}{1 + .05 \times 0.1} = 10.13.$$

Further,

$$c = \frac{39.35 \times 0.50898 + 10.13 \times (1 - 0.50898)}{1 + .05 \times 0.1} = 24.88.$$

- Thus
$$\Delta = \frac{c_u - c_d}{S_u - S_d} = \frac{39.35 - 10.13}{248.92 - 212.52} = 0.8027.$$
- Further,
$$\Gamma_2 = \frac{c_{uu} - c_{ud}}{S_{uu} - S} = \frac{58.40 - 20}{268.40 - 230} = 1.$$
and
$$\Gamma_1 = \frac{c_{ud} - c_{dd}}{S - S_{dd}} = \frac{20}{230 - 196.36} = 0.5945,$$
also $h = (268.40 - 196.36)/2 = 36.02$. Thus $\Gamma = (1 - 0.5945)/36.02 = 0.0113$.
- Moreover,
$$\Theta = \frac{c_{ud} - c}{2\Delta t} = \frac{20 - 24.88}{2 \times 0.1} = -24.4(Annual).$$
Alternatively, the daily theta is $\Theta = -24.4/365 = -0.066849315$. This mean each passing day the value of option go down by 0.0668 approximately assuming everything else is same.

# Bermudan options

# Bermudan Option

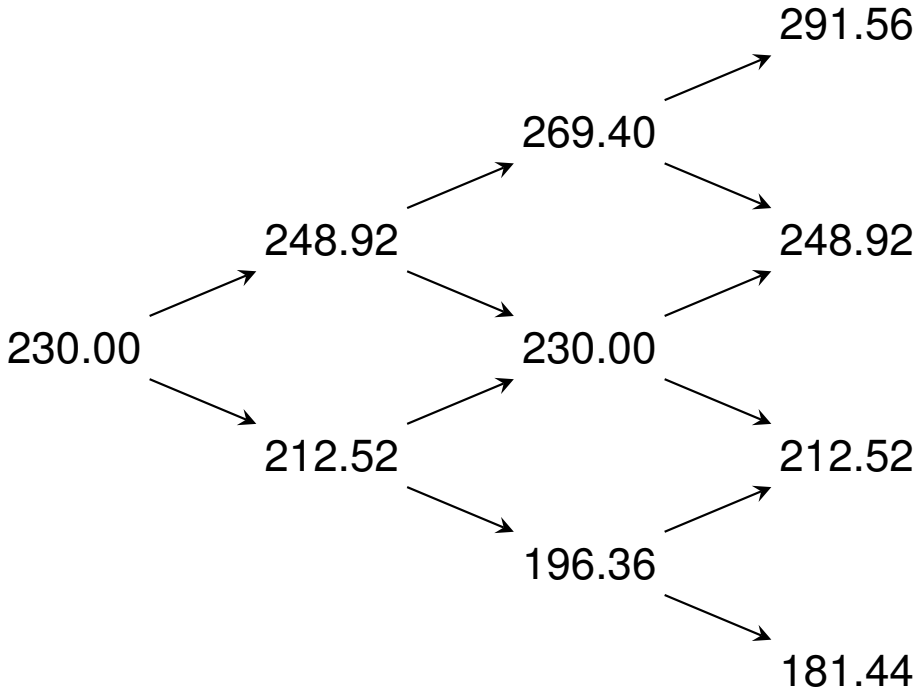
- Bermudan options take an intermediate place between American and European options.
- American options can be exercised at any time, while, the European can be exercised only at maturity.
- A Bermudan option has fixed set of dates at which the option can be exercised, e.g., annually, quarterly, or monthly.

**Table:** Bermudan put option features

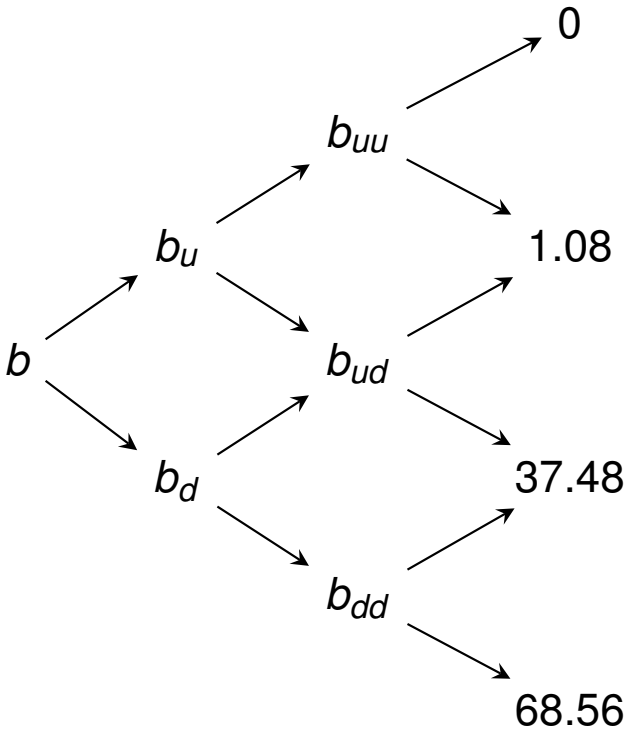
| | | |
|---|---|---|
| Current stock price | $S$ | 230.00 |
| Exercise price | $K$ | 250.00 |
| Time to maturity | $T$ | 0.30 |
| Volatility | $\sigma$ | 0.25 |
| Risk free rate | $r$ | 0.05 |
| Time steps | $n$ | 3 |
| Exercise Times | | At $t = 0.10, 0.30$ |
| Option type | | Bermudan put |

- We have $\Delta t = T/n = 0.1$, $u = e^{\sigma\sqrt{\Delta}} = e^{0.25\sqrt{0.1}} = 1.0823$, and $d = 1/u = 0.9234$.
- Given the current stock price $S = 230$ the stock can either increase to $S_u = uS = 248.92$ or decrease to $S_d = dS = S/u = 212.52$.
- Further $\tilde{p} = \frac{e^{0.05\times 0.1} - 0.9234}{1.0823 - 0.9234} = \frac{1.005013 - 0.9234}{1.0823 - 0.9234} \approx 0.512$.

# Equity price evolution under CRR model



# Put option Payoffs



At $t = 0.2$, we price like European option

$$b_{uu} = e^{-0.05\times 0.1}(0.512 \times 0 + 0.488 \times 1.08) = 0.5244;$$
$$b_{ud} = e^{-0.05\times 0.1}(0.512 \times 1.08 + 0.488 \times 37.48) = 18.7492;$$
$$b_{dd} = e^{-0.05\times 0.1}(0.512 \times 37.48 + 0.488 \times 68.56) = 52.3845.$$

At $t = 0.1$, it is priced like American style

$$\begin{aligned} b_u &= \max\{1.08, e^{-0.05\times 0.1}(0.512 \times 0.5244 + 0.488 \times 18.7492)\} \\ &= \max\{1.08, 9.3711\} = 9.3711; \\ b_d &= \max\{37.48, e^{-0.05\times 0.1}(0.512 \times 18.7492 + 0.488 \times 52.3845)\} \\ &= \max\{37.48, 34.9878\} = 37.48. \end{aligned}$$

Thus

$$b = e^{-0.05\times 0.1}(0.512 \times 9.3711 + 0.488 \times 37.48) = 22.9731.$$

# Barrier options

# Barrier Options

- **Barrier** option payoffs depend on whether the underlying hits a certain level before expiration date. The level is called the barrier.
- The barrier options can be **knock-in** or **knock-out**.
- A knock-in option is worthless until the underlying touches the barrier price.
- A knock-out option become worthless if exceeds the barrier price.

# Barrier option types

The four main types of barrier options are

- **Up-and-out:** The spot price starts below the barrier level.
- **Down-and-out:** The spot price starts above the barrier level.
- **Up-and-in:** The spot price starts below the barrier level and has to move up for the option to become worthy.
- **Down-and-in:** The spot price starts above the barrier level and has to move down for the option to become worthy.
- You can have a call or put with American, European or Bermudan exercise style.

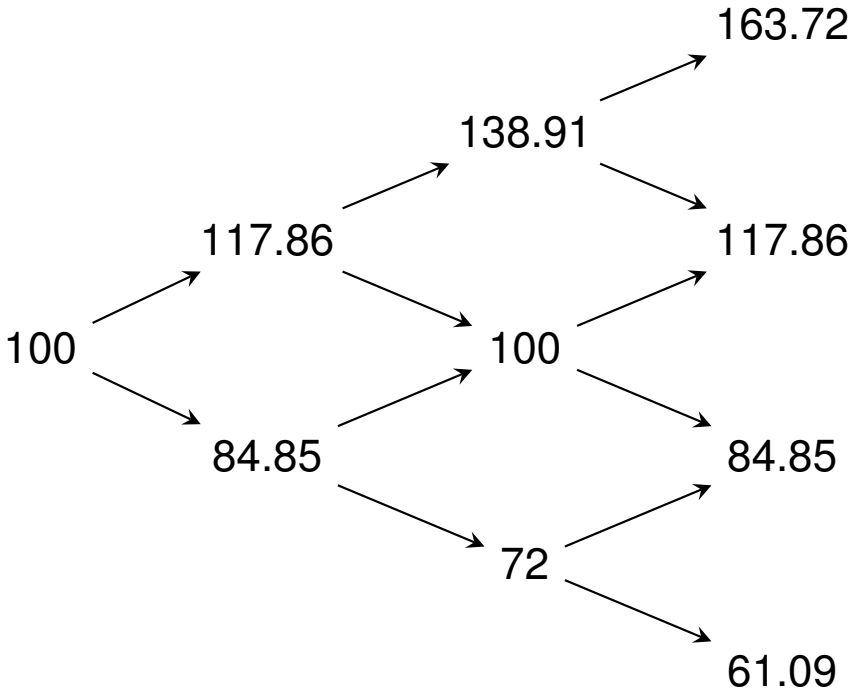
# European call for an up-and-out barrier option

**Table:** Barrier option features

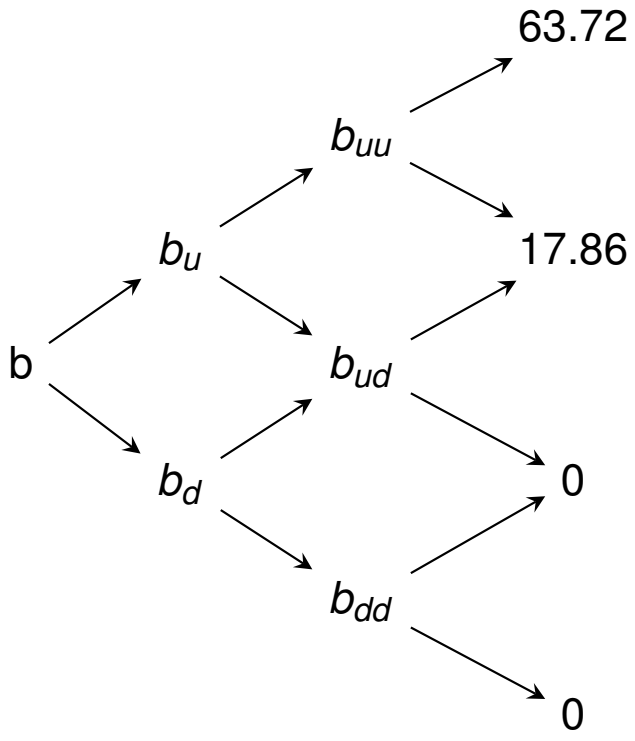
| | | |
|---|---|---|
| Current stock price | $S$ | 100 |
| Exercise price | $K$ | 100 |
| Time to maturity | $T$ | 0.9 |
| Volatility | $\sigma$ | 0.30 |
| Risk free rate | $r$ | 0.05 |
| Barrier | $B$ | 130 |
| Number of steps | $n$ | 3 |
| Size of steps | $\Delta t$ | 0.3 |
| Option type | | up-and-out barrier option |

We have $u = 1.1786$, $d = 0.8485$ and $\tilde{p} = 0.5048$.

# Equity price evolution under CRR model



# Payoff at maturity




We have the price of the barrier option is
$b = \exp(-0.05 \times 0.9)2\tilde{p}^2(1-\tilde{p}) \times 17.86 = 4.31$. The value of the vanilla European option is

$$c = \exp(-0.05 \times 0.9)(\tilde{p}^3 \times 63.72 + 3\tilde{p}^2(1-\tilde{p}) \times 17.86 + 0 + 0) = 14.30.$$

# Asian options

# Asian Options

- **Asian options** are a particular type of exotic options that are priced based on the average price of the underlying instrument.
- The Asian options can be exercised in European or American style.
- Asian options are typically cheaper than European or American options.
- Payoff of Asian call with arithmetic averaging is

$$Payoff = \max\left\{\frac{\sum_{i=1}^{n} S_i}{n} - K, 0\right\}.$$

- Payoff of Asian call with geometric averaging is

$$Payoff = \max\left\{\left(\prod_{i=1}^{n} S_i\right)^{1/n} - K, 0\right\}.$$

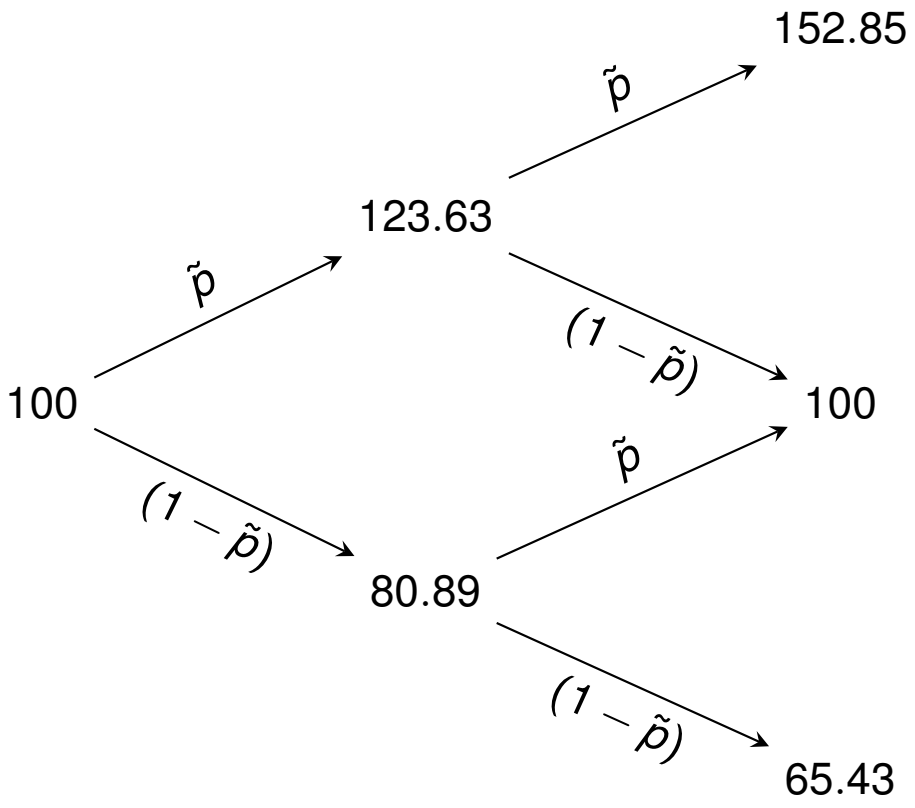
# Asian call with arithmetic averaging

**Table:** Asian option features

| | | |
|---|---|---|
| Current stock price | $S$ | 100 |
| Exercise price | $K$ | 100 |
| Time to maturity | $T$ | 1 |
| Volatility | $\sigma$ | 0.30 |
| Risk free rate | $r$ | 0.08 |
| Number of steps | $n$ | 2 |
| Size of steps | $\Delta t$ | 0.5 |
| Option type | | Asian call |

We have $u = 1.2363$, $d = 0.8089$ and $\tilde{p} = 0.5426$.

# Underlying price



# Asian Option Price

| | t=0 | t=0.5 | t=1 | Average | Probability | Payoff | | |
|---|---|---|---|---|---|---|---|---|
| | 100 | 123.63 | 152.85 | 125.4933 | 0.294415 | 25.49333 | | |
| | 100 | 123.63 | 100 | 107.8767 | 0.248185 | 7.876667 | | |
| | 100 | 80.89 | 100 | 93.63 | 0.248185 | 0 | | |
| | 100 | 80.89 | 65.43 | 82.10667 | 0.209215 | 0 | | |
| | | | | | | | | |
| | | | | Expected Payoff | | 9.460486 | | |
| | | | | Discounted Value | | 8.733129 | | |
| | | | | | | | | |
| | | | | | | | | |

# Some References


- Papanicolaou, A. (2017). *Introduction to Stochastic Differential Equations (SDEs) for Finance.* arXiv:1504.05309v13
- Hull, J. C. (2011). *Options, Futures and Other Derivatives*. 8th ed., Prentice Hall.
- Shreve, S. (2005). *Stochastic Calculus for Finance I: The Binomial Asset Pricing Model*. Springer.
- Boyle, P. and McDougall, J. (2005). *Trading and Pricing Financial Derivatives*. Walter De Gruyter, Berlin.

**THANK YOU**